Effect of CT Reconstruction Filters on AI Lung Tumor Volumetry and Mitigation with Radiologist Box Prompts

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Abstract

Lung cancer screening increasingly relies on tumor volume to judge nodule growth, and artificial intelligence (AI) is often used to measure it. AI segmentation, however, is typically validated with Dice overlap, and the same CT acquisition can be reconstructed with different filters chosen by each site. We used the National Lung Screening Trial as a natural experiment: among 575 participants with expert tumor masks from four scanner vendors, 128 had the identical acquisition archived with two reconstruction filters. A 2.5D U-Net (three seeds) and nnU-Net segmented both reconstructions, which were compared with the same expert mask under a protocol fixed before training, with the test set scored once. Reconstruction did not change Dice (difference −0.009; 95% CI −0.027 to 0.007; p=0.49), but it changed AI tumor volume by a median of 9.6% and by more than 25%, a common volumetric growth threshold, in 24% of patients (95% CI 17–32%); nnU-Net showed the same effect in 15% (10–22%). A model guided by one radiologist-drawn box per tumor raised test Dice from 0.63 to 0.75 (n=71; $p<10^{-7}$), comparable to nnU-Net without a box (0.72), and, in an exploratory analysis, reduced reconstruction-induced volume changes above 25% from 25% to 3% of patients. In post-hoc analyses, its repeatability coefficient across reconstructions was 21%, versus 91% for nnU-Net and 194% for the U-Net. Dice alone can therefore certify an AI tool whose volume measurements shift with a routine scanner setting. Volume stability across reconstructions should be evaluated before clinical deployment, and a single radiologist box prompt offers a practical safeguard.